ELECTROMAGNETIC THEORY · EXAM STUDY GUIDE

Electrostatic Energy & Laplace's Equation

From first principles to exam-level problems. Concepts, interpretations, every key formula, and 22 fully worked practice questions.

Part A

Energy of charge systems · energy density $w_e = \frac{1}{2}\epsilon E^2$ · capacitor energy · forces by virtual displacement

Part B

Poisson $\nabla^2 V = -\rho/\epsilon$ · Laplace $\nabla^2 V = 0$ · uniqueness · 1-D solutions · separation of variables

Practice

22 exam-style questions with step-by-step, hand-held solutions · 24 rapid-fire concept checks

Notation follows D. K. Cheng, *Field and Wave Electromagnetics*, §3-11 and Chapter 4

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1 How to use this guide in the next 4 hours

Time	What to do
0:00 – 0:20	Part 0 (basics). Don't skip it. Everything else uses $\mathbf{E} = -\nabla V$, $\nabla \cdot \mathbf{D} = \rho$ and $\rho_s = \epsilon E_n$.
0:20 – 1:10	Part A and A2 (energy and forces). Then attempt Q1 to Q10 *before* reading the solutions.
1:10 – 2:20	Part B to B3 (Poisson, Laplace, uniqueness, 1-D solutions). Then attempt Q11 to Q19.
2:20 – 2:50	Part B4 (separation of variables) and Q20 to Q22. If you are short on time, read only the method box and the final answers.
2:50 – 3:30	Write the formula sheet out by hand from memory twice. Writing it out is what makes it stick.
3:30 – 4:00	Rapid-fire checks, then rest.

2 Part 0: The basics you need first

2.1 The cast of characters

Symbol	Name	Unit	What it means in plain words
Q, q	Charge	C (coulomb)	The "source" of everything in electrostatics.
ρ (or ρ_v)	Volume charge density	C/m³	Charge per unit volume.
ρ_s	Surface charge density	C/m²	Charge per unit area (sits on conductor surfaces).
ρ_ℓ	Line charge density	C/m	Charge per unit length.
$\mathbf{E}$	Electric field intensity	V/m (= N/C)	Force per unit positive charge. Points from + to −, i.e. from high V to low V.
V	Electric potential	V (volt = J/C)	Work per unit charge to bring a test charge from infinity (zero reference) to the point.
$\mathbf{D}$	Electric flux density (displacement)	C/m²	$\mathbf{D} = \epsilon\mathbf{E}$. Depends only on free charges.
$\epsilon = \epsilon_0\epsilon_r$	Permittivity	F/m	How a material responds to E. $\epsilon_0 = 8.854 \times 10^{-12} \approx \frac{10^{-9}}{36\pi}$ F/m.
C	Capacitance	F (farad)	$C = Q/V$: charge stored per volt.
W_e	Electrostatic energy	J	Work done to assemble the charges = energy stored.
w_e	Energy density	J/m³	Energy per unit volume of the field.

Useful constant: $\dfrac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \approx 9 \times 10^9 \ \mathrm{m/F}$.

2.2 The four relations you'll use constantly

MEMORISE

(1) $\mathbf{E} = -\nabla V$ **(2)** $\nabla \cdot \mathbf{D} = \rho$ **(3)** $\mathbf{D} = \epsilon\mathbf{E}$ **(4)** $\rho_s = \mathbf{a}_n \cdot \mathbf{D} = \epsilon E_n$ (on a conductor surface)

- (1) says the field points "downhill" in potential. The minus sign matters.
- (2) is Gauss's law in point form: charge is where D "spreads out".
- (4) To find the charge sitting on a conductor, take the normal component of D pointing *out of the conductor into the field region.*

2.3 Gradient (for $\mathbf{E} = -\nabla V$) in the three coordinate systems

$$\text{Cartesian: } \nabla V = \mathbf{a}_x \frac{\partial V}{\partial x} + \mathbf{a}_y \frac{\partial V}{\partial y} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\text{Cylindrical } (r, \phi, z): \ \nabla V = \mathbf{a}_r \frac{\partial V}{\partial r} + \mathbf{a}_\phi \frac{1}{r} \frac{\partial V}{\partial \phi} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\text{Spherical } (R, \theta, \phi): \ \nabla V = \mathbf{a}_R \frac{\partial V}{\partial R} + \mathbf{a}_\theta \frac{1}{R} \frac{\partial V}{\partial \theta} + \mathbf{a}_\phi \frac{1}{R \sin\theta} \frac{\partial V}{\partial \phi}$$

TRAP

The $\frac{1}{r}$, $\frac{1}{R}$ and $\frac{1}{R\sin\theta}$ factors on the angular terms are the most common lost marks. An angle has no units, so dividing by a length is what turns "volts per radian" into V/m.

2.4 Standard capacitances (they appear in both topics)

Geometry	Capacitance
Parallel plate, area S, gap d	$C = \dfrac{\epsilon S}{d}$
Coaxial (cylindrical), radii $a < b$, length L	$C = \dfrac{2\pi\epsilon L}{\ln(b/a)}$
Spherical (concentric), radii $a < b$	$C = \dfrac{4\pi\epsilon}{\frac{1}{a} - \frac{1}{b}} = \dfrac{4\pi\epsilon ab}{b - a}$
Isolated sphere of radius a (let $b \to \infty$)	$C = 4\pi\epsilon a$

3 Part A: Electrostatic energy and energy density

3.1 The core idea: energy = work done to assemble the charges

Charges push or pull each other, so bringing charges together takes work. Electrostatic fields are **conservative**: the work done doesn't depend on the path, and it isn't lost. It is **stored as potential energy** in the arrangement. That stored energy is W_e (the subscript e stands for "electric").

The bookkeeping uses the definition of potential: *V at a point = work per unit charge to bring a charge from infinity to that point*. So bringing charge Q to a place where the potential is V costs work QV.

3.2 Building it up one charge at a time

1. **First charge Q_1.** Space is empty, so there is no field to fight. Work = 0.
2. **Second charge Q_2**, brought to distance R_{12} from Q_1. It moves against the field of Q_1:

$$W_2 = Q_2 V_2 = Q_2 \frac{Q_1}{4\pi\epsilon_0 R_{12}} \qquad \text{(Cheng 3-159)}$$

   The expression is symmetric, so it is also $Q_1 V_1$. Averaging the two gives $W_2 = \frac{1}{2}(Q_1 V_1 + Q_2 V_2)$.

3. **Third charge Q_3** must fight both:

$$\Delta W = Q_3 \left(\frac{Q_1}{4\pi\epsilon_0 R_{13}} + \frac{Q_2}{4\pi\epsilon_0 R_{23}} \right)$$

$$W_3 = \frac{1}{4\pi\epsilon_0}\left(\frac{Q_1 Q_2}{R_{12}} + \frac{Q_1 Q_3}{R_{13}} + \frac{Q_2 Q_3}{R_{23}} \right) = \tfrac{1}{2}(Q_1 V_1 + Q_2 V_2 + Q_3 V_3)$$

FORMULA: N DISCRETE POINT CHARGES (CHENG 3-165)

$$W_e = \frac{1}{2}\sum_{k=1}^{N} Q_k V_k \quad \text{(J)}, \qquad V_k = \frac{1}{4\pi\epsilon_0}\sum_{j \neq k} \frac{Q_j}{R_{jk}}$$

Equivalent "sum over pairs" form (easier for numbers; every pair counted **once**):

$$W_e = \frac{1}{4\pi\epsilon_0} \sum_{\text{all pairs } i<j} \frac{Q_i Q_j}{R_{ij}}$$

INTERPRETATION: WHY THE ½?

In $\sum Q_k V_k$ every pair is counted twice: once as "Q_1 sitting in Q_2's potential" and once as "Q_2 sitting in Q_1's potential". The ½ removes the double counting.

TWO REMARKS FROM THE TEXTBOOK (OFTEN ASKED AS THEORY)

1. W_e **can be negative.** With opposite charges the field does the work: the charges attract and you hold them back as they come in. A negative total energy means a bound system.
2. **The discrete formula gives only the interaction (mutual) energy.** It leaves out the *self-energy*, the work needed to build each point charge itself. For a true point charge the self-energy is infinite, since it behaves like $Q^2/(8\pi\epsilon_0 a)$ with $a \to 0$.

Electron-volt: $1\ \text{eV} = 1.60 \times 10^{-19}\ \text{J}$. It is the energy gained by an electron moving through a potential difference of 1 V. To convert J to eV, divide by 1.6×10^{-19}.

3.3 Continuous charge distributions

Replace $Q_k \to \rho\, dv$ and the sum by an integral:

FORMULA (CHENG 3-170)

$$W_e = \frac{1}{2}\int_{V'} \rho V\, dv \quad \text{(J)}$$

Here V is the potential at the location of $\rho\, dv$. **This version includes self-energy**, because every infinitesimal piece interacts with every other piece. Analogous forms: $\frac{1}{2}\oint \rho_s V\, ds$ for surface charge and $\frac{1}{2}\int \rho_\ell V\, d\ell$ for line charge.

Special case: a charged conductor. The whole conductor is at one potential V, so $W_e = \frac{1}{2}V\oint \rho_s\, ds = \frac{1}{2}QV$.

3.4 Energy in terms of the FIELD (the most important form)

Often we know $\mathbf{E}$ but not ρ. Derivation (worth knowing, as it is sometimes asked):

1. Put $\rho = \nabla\cdot\mathbf{D}$ into $W_e = \frac{1}{2}\int \rho V\, dv$: $W_e = \frac{1}{2}\int (\nabla\cdot\mathbf{D})V\, dv$.
2. Vector identity: $\nabla\cdot(V\mathbf{D}) = V\nabla\cdot\mathbf{D} + \mathbf{D}\cdot\nabla V$, so $(\nabla\cdot\mathbf{D})V = \nabla\cdot(V\mathbf{D}) - \mathbf{D}\cdot\nabla V$.
3. Divergence theorem on the first term: $W_e = \frac{1}{2}\oint_{S'} V\mathbf{D}\cdot\mathbf{a}_n\, ds + \frac{1}{2}\int \mathbf{D}\cdot\mathbf{E}\, dv$ (using $-\nabla V = \mathbf{E}$).
4. Let the surface be a huge sphere of radius $R \to \infty$. Then $V \sim 1/R$, $D \sim 1/R^2$ and area $\sim R^2$, so the surface term $\sim 1/R \to 0$. What remains is:

FORMULA (CHENG 3-176 A, B, C)

$$W_e = \frac{1}{2}\int_{V'} \mathbf{D}\cdot\mathbf{E}\, dv = \frac{1}{2}\int_{V'} \epsilon E^2\, dv = \frac{1}{2}\int_{V'} \frac{D^2}{\epsilon}\, dv \qquad \text{(J)}$$

3.5 Energy density w_e

Define w_e so that $W_e = \int_{V'} w_e\, dv$. From the above:

FORMULA: ENERGY DENSITY (CHENG 3-178 A, B, C)

$$\boxed{w_e = \tfrac{1}{2}\mathbf{D}\cdot\mathbf{E} = \tfrac{1}{2}\epsilon E^2 = \frac{D^2}{2\epsilon} \quad (\mathrm{J/m^3})}$$

INTERPRETATIONS YOU SHOULD BE ABLE TO WRITE IN WORDS

- Energy is pictured as **stored in the field itself**, spread through space, with the most energy where $\boldsymbol{E}$ is strongest.
- $w_e \propto E^2$: doubling the field quadruples the energy density.
- **For a given E**, a higher ϵ stores more energy ($\frac{1}{2}\epsilon E^2$). **For a given D**, a higher ϵ stores less ($D^2/2\epsilon$). Which form to use depends on what is held fixed.
- Cheng's caveat: this definition is a mathematical convenience. No experiment can locate energy at a particular point. Only the total integral W_e is physically guaranteed.
- Units check: ϵE^2 has units (F/m)(V/m)² = F·V²/m³ = J/m³, since F·V² = C·V = J.
- Energy density is also a **pressure**: $\frac{1}{2}\epsilon E^2$ is the outward force per unit area (N/m²) on a conductor surface. See Part A2.

3.6 Capacitor energy

Parallel plates (Cheng Ex. 3-24): $E = V/d$ is uniform, and the volume is Sd:

$$W_e = \tfrac{1}{2}\epsilon\left(\frac{V}{d}\right)^2 (Sd) = \tfrac{1}{2}\left(\frac{\epsilon S}{d}\right)V^2$$

FORMULA: TRUE FOR ANY TWO-CONDUCTOR CAPACITOR (CHENG 3-180)

$$W_e = \tfrac{1}{2}CV^2 = \tfrac{1}{2}QV = \frac{Q^2}{2C} \quad (\mathrm{J})$$

EXAM TRICK: CAPACITANCE FROM ENERGY (CHENG EX. 3-25)

1. Assume charge Q and find $\mathbf{E}$ by Gauss's law.
2. Compute $W_e = \frac{1}{2}\int \epsilon E^2\, dv$.
3. Set it equal to $Q^2/2C$ and solve: $C = \dfrac{Q^2}{2W_e}$ (or $C = 2W_e/V^2$).

For coax this gives $W_e = \dfrac{Q^2}{4\pi\epsilon L}\ln\dfrac{b}{a} \Rightarrow C = \dfrac{2\pi\epsilon L}{\ln(b/a)}$.

3.7 Classic results to recognise on sight

System	Stored energy
Uniformly charged sphere, radius b, total charge Q (self-energy)	$W_e = \dfrac{4\pi\rho^2 b^5}{15\epsilon_0} = \dfrac{3Q^2}{20\pi\epsilon_0 b}$

Conducting sphere (or spherical shell) of radius a, charge Q	$W_e = \dfrac{Q^2}{8\pi\epsilon_0 a}$
Coaxial capacitor, length L	$W_e = \dfrac{Q^2}{4\pi\epsilon L}\ln\dfrac{b}{a}$
Spherical capacitor	$W_e = \dfrac{Q^2}{8\pi\epsilon}\left(\dfrac{1}{a}-\dfrac{1}{b}\right)$
Parallel plate	$W_e = \frac{1}{2}\dfrac{\epsilon S}{d}V^2 = \dfrac{Q^2 d}{2\epsilon S}$

3.8 Battery connected vs disconnected (a favourite conceptual question)

Change	Battery disconnected (Q constant): use $W = Q^2/2C$	Battery connected (V constant): use $W = \frac{1}{2}CV^2$
Double the gap d ($C \to C/2$)	W **doubles**. E unchanged ($E = \rho_s/\epsilon$), V doubles.	W **halves**. E halves, Q halves.
Fill with dielectric ϵ_r ($C \to \epsilon_r C$)	W becomes W/ϵ_r (decreases). The dielectric is pulled in.	W becomes $\epsilon_r W$ (increases). The battery supplies $2\times$ that extra energy.

4 Part A2: Electrostatic forces from energy (virtual displacement)

Finding forces with Coulomb's law is tedious for conductors. Instead: *imagine* moving a body by a tiny distance $d\boldsymbol{\ell}$ (a "virtual displacement") and see how the energy changes.

Case 1: Fixed charges (isolated system, no battery)

The mechanical work done by the field must come out of the stored energy: $\mathbf{F}_Q \cdot d\boldsymbol{\ell} = -dW_e = -(\nabla W_e) \cdot d\boldsymbol{\ell}$.

FORMULA (CHENG 3-185)

$$\mathbf{F}_Q = -\nabla W_e \quad (\mathrm{N}) \qquad \text{e.g. } (F_Q)_x = -\frac{\partial W_e}{\partial x}$$

Case 2: Fixed potentials (battery connected)

Now the battery supplies energy $dW_s = \sum V_k \, dQ_k$. Half of it goes into stored energy and half into mechanical work: $dW_e = \frac{1}{2} dW_s$, and so $dW = dW_e$.

FORMULA

$$\mathbf{F}_V = +\nabla W_e \quad (\mathrm{N})$$

INTERPRETATION

Both formulas give the **same physical force**. The sign differs only because W_e is written in terms of different held-constant quantities (Q in one case, V in the other). Bodies move so as to *decrease* W_e when charge is fixed and to *increase* W_e when potential is fixed. Either way, they move so as to **increase capacitance**.

Force between capacitor plates

$$|F| = \frac{Q^2}{2\epsilon S} = \frac{\epsilon S V^2}{2d^2} = \tfrac{1}{2}\epsilon E^2 \, S = \frac{\rho_s^2}{2\epsilon} S \qquad \text{(attractive)}$$

ELECTROSTATIC PRESSURE ON A CONDUCTOR SURFACE

$$p = \frac{F}{S} = \frac{\rho_s^2}{2\epsilon} = \tfrac{1}{2}\epsilon E^2 = w_e \quad (\mathrm{N/m^2})$$

The force is always directed **outward from the conductor**, into the field region. For a capacitor that means each plate is pulled toward the other.

Torque version for rotation: $(T_Q)_z = -\dfrac{\partial W_e}{\partial \phi}$ (fixed charge), $(T_V)_z = +\dfrac{\partial W_e}{\partial \phi}$ (fixed V).

5 Part B: Poisson's and Laplace's equations

5.1 Why do we need them?

In Chapter 3 you found V and E *from a known charge distribution*. In real devices (capacitors, cables, electrodes), you usually know the **potentials of the conductors** ("this plate is at 100 V, that one is grounded"), but you don't know how charge distributes itself on them. That's a **boundary-value problem**: solve a differential equation in the region between conductors, subject to the known boundary potentials.

5.2 Derivation (3 lines; learn it)

1. Gauss: $\nabla \cdot \mathbf{D} = \rho$, with $\mathbf{D} = \epsilon \mathbf{E}$, so $\nabla \cdot (\epsilon \mathbf{E}) = \rho$.
2. Irrotational field: $\nabla \times \mathbf{E} = 0$, so $\mathbf{E} = -\nabla V$. Hence $\nabla \cdot (\epsilon \nabla V) = -\rho$.
3. Simple medium (linear, isotropic, homogeneous: ϵ constant) lets ϵ come out:

POISSON'S EQUATION (CHENG 4-6)

$$\nabla^2 V = -\frac{\rho}{\epsilon} \qquad (\mathrm{V/m^2})$$

LAPLACE'S EQUATION (CHENG 4-10): CHARGE-FREE REGION, $\rho = 0$

$$\nabla^2 V = 0$$

∇^2 ("del squared", the **Laplacian**) means "divergence of the gradient": $\nabla^2 V = \nabla \cdot \nabla V$. It's a second-order PDE.

PHYSICAL MEANING OF THE LAPLACIAN

$\nabla^2 V$ at a point measures how much V there differs from the *average* of V in its immediate neighbourhood. $\nabla^2 V = 0$ means "V at each point equals the average of its surroundings": no bumps, no dips. Charge is what creates bumps or dips. Positive ρ gives $\nabla^2 V < 0$, a local hill in V.

5.3 The Laplacian in all three coordinate systems

MEMORISE (CHENG 4-7, 4-8, 4-9)

$$\text{Cartesian: } \nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{Cylindrical: } \nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{Spherical: } \nabla^2 V = \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial V}{\partial R}\right) + \frac{1}{R^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial V}{\partial \theta}\right) + \frac{1}{R^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2}$$

MEMORY AID

Radial parts: cylindrical has r *inside and outside* ($\frac{1}{r}\,\partial_r\, r\,\partial_r$). Spherical has R^2 inside and outside ($\frac{1}{R^2}\,\partial_R\, R^2\,\partial_R$). The θ term mirrors that pattern with $\sin\theta$.

5.4 Terms and their meanings

Term	Meaning
Simple medium	Linear, isotropic and homogeneous, so ϵ is a constant scalar. Needed to pull ϵ out of the divergence.
Harmonic function	Any function satisfying Laplace's equation.
Boundary-value problem	Solve a PDE in a region given conditions on its boundary.
Dirichlet boundary condition	V is specified on the boundary (e.g. conductor at V_0).
Neumann boundary condition	$\partial V/\partial n$ is specified on the boundary, i.e. normal E, i.e. surface charge.
Mixed boundary condition	V on part of the boundary, $\partial V/\partial n$ on the rest.
Fringing	Field bending at a capacitor's edges. We usually neglect it, so V depends on one coordinate only.
Space-charge limited	Vacuum diode where the field at the cathode is zero ($dV/dy = 0$ at $y = 0$). Gives Child–Langmuir $J \propto V_0^{3/2}$.

5.5 Laplace vs Poisson: which one?

DECISION RULE

- Region between conductors with **no charge** (vacuum or ideal dielectric): **Laplace**.
- Region containing **volume charge** (electron cloud, space charge in a diode, a p-n junction): **Poisson**.
- Surface charges on conductors are *not* volume charge. They live on the boundary and are handled through boundary conditions. So a capacitor is a Laplace problem even though its plates are charged.

6 Part B2: Uniqueness theorem and properties of harmonic functions

UNIQUENESS THEOREM (CHENG 4-3)

*A solution of Poisson's equation (Laplace's equation is the special case $\rho = 0$) that satisfies the given boundary conditions is the **only** solution.*

Proof idea (sometimes asked): suppose V_1 and V_2 both work. Let $V_d = V_1 - V_2$. Then $\nabla^2 V_d = 0$ inside and $V_d = 0$ on the conductors. Using the identity $\nabla \cdot (V_d \nabla V_d) = V_d \nabla^2 V_d + |\nabla V_d|^2$ and the divergence theorem, $\oint V_d \nabla V_d \cdot d\mathbf{s} = \int |\nabla V_d|^2 dv$. The left side is zero because $V_d = 0$ on the boundary, and also at infinity. So $\int |\nabla V_d|^2 dv = 0$, which forces $\nabla V_d = 0$ everywhere. V_d is then a constant, and that constant is 0 because $V_d = 0$ on the boundary. So $V_1 = V_2$.

WHY IT MATTERS (INTERPRETATION)

- You may **guess** a solution by any method: inspection, symmetry, images, separation of variables. If it satisfies the equation and the boundary conditions, it is THE answer.
- This is the justification for the **method of images**.

Properties of solutions of Laplace's equation

1. **Mean-value property**: V at a point = average of V over any sphere (or circle, in 2-D) centred there that encloses no charge.
2. **No local maxima or minima** inside a charge-free region. The extremes of V occur on the boundaries (conductors).
3. Consequence, **Earnshaw's theorem**: a charge can't be held in stable equilibrium by electrostatic forces alone.
4. **Superposition**: Laplace's equation is linear, so if V_1 and V_2 are solutions, so is $aV_1 + bV_2$. (This is used in separation of variables.)
5. Laplace's equation alone gives nothing until boundary conditions are imposed. The boundary conditions fix the integration constants.

7 Part B3: Solving Laplace's equation, the 1-D cases

THE UNIVERSAL 6-STEP RECIPE (USE THIS FOR EVERY PROBLEM)

1. Pick the coordinate system that matches the conductor shapes: planes → Cartesian, coaxial cylinders or wedges → cylindrical, concentric spheres or cones → spherical.
2. Decide which ONE coordinate V depends on, by symmetry and by neglecting fringing. Drop all derivatives with respect to the others.
3. Write the reduced ODE and integrate twice. This gives two constants C_1, C_2.
4. Apply the two boundary conditions (the two conductor potentials) to find C_1 and C_2.
5. $\mathbf{E} = -\nabla V$ (mind the scale factors), then $\mathbf{D} = \epsilon \mathbf{E}$.
6. $\rho_s = \mathbf{a}_n \cdot \mathbf{D}$ on each conductor ($\mathbf{a}_n$ points away from the conductor). Then $Q = \int \rho_s \, ds$, and $C = Q/V_0$.

7.1 The five standard 1-D solutions

Geometry	V depends on	Reduced equation	General solution
Parallel plates	y (or x)	$\frac{d^2V}{dy^2} = 0$	$V = C_1 y + C_2$
Coaxial cylinders	r	$\frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0$	$V = C_1 \ln r + C_2$
Wedge: two half-planes meeting at an angle	ϕ	$\frac{1}{r^2}\frac{d^2V}{d\phi^2} = 0$	$V = C_1 \phi + C_2$
Concentric spheres	R	$\frac{1}{R^2}\frac{d}{dR}\left(R^2\frac{dV}{dR}\right) = 0$	$V = -\frac{C_1}{R} + C_2$
Cone vs plane (coaxial cones)	θ	$\frac{1}{R^2 \sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dV}{d\theta}\right) = 0$	$V = C_1 \ln\left(\tan\frac{\theta}{2}\right) + C_2$

How each is integrated (so you can reproduce it)

- **Cylindrical r:** $r\,dV/dr = C_1 \Rightarrow dV/dr = C_1/r \Rightarrow V = C_1 \ln r + C_2$.
- **Spherical R:** $R^2\,dV/dR = C_1 \Rightarrow dV/dR = C_1/R^2 \Rightarrow V = -C_1/R + C_2$.
- **Spherical θ:** $\sin\theta\,dV/d\theta = C_1 \Rightarrow dV/d\theta = C_1/\sin\theta \Rightarrow V = C_1 \ln\tan(\theta/2) + C_2$, using $\int \csc\theta\,d\theta = \ln\tan(\theta/2)$.

7.2 The fully solved results (from these, every capacitance falls out)

Problem	V	E	C
Plates: $V=0$ at $y=0$, V_0 at $y=d$	$V_0\dfrac{y}{d}$	$-\mathbf{a}_y\dfrac{V_0}{d}$	$\dfrac{\epsilon S}{d}$
Coax: V_0 at $r=a$, 0 at $r=b$	$V_0\dfrac{\ln(b/r)}{\ln(b/a)}$	$\mathbf{a}_r\dfrac{V_0}{r\ln(b/a)}$	$\dfrac{2\pi\epsilon L}{\ln(b/a)}$
Spheres: V_0 at $R=a$, 0 at $R=b$	$V_0\dfrac{\frac{1}{R}-\frac{1}{b}}{\frac{1}{a}-\frac{1}{b}}$	$\mathbf{a}_R\dfrac{V_0}{R^2\left(\frac{1}{a}-\frac{1}{b}\right)}$	$\dfrac{4\pi\epsilon}{\frac{1}{a}-\frac{1}{b}}$
Wedge: 0 at $\phi=0$, V_0 at $\phi=\alpha$	$V_0\dfrac{\phi}{\alpha}$	$-\mathbf{a}_\phi\dfrac{V_0}{\alpha r}$	–
Cone $\theta=\alpha$ at V_0, plane $\theta=\pi/2$ at 0	$V_0\dfrac{\ln\tan(\theta/2)}{\ln\tan(\alpha/2)}$	$-\mathbf{a}_\theta\dfrac{V_0}{R\sin\theta\,\ln\tan(\alpha/2)}$	–

INTERPRETATION

Only between parallel plates is V linear and E uniform. In a coax, E ∝ 1/r is strongest at the inner conductor; in spheres, E ∝ 1/R² is also strongest at the inner one. That is where dielectric breakdown begins. Inside a wedge, E ∝ 1/r grows toward the edge (the field concentrates at sharp edges).

7.3 Poisson examples (when charge is present)

(a) Uniform charge ρ_0 between grounded plates at $y=0$ and $y=d$:
$\dfrac{d^2V}{dy^2}=-\dfrac{\rho_0}{\epsilon}\Rightarrow V=-\dfrac{\rho_0}{2\epsilon}y^2+C_1y+C_2$. With $V(0)=V(d)=0$: $V=\dfrac{\rho_0}{2\epsilon}(yd-y^2)$, a parabola with maximum $\rho_0d^2/8\epsilon$ at the centre. (Q15 below.)

(b) Vacuum diode, space-charge limited (Cheng Ex. 4-2): cathode at $y=0$ (V=0), anode at $y=d$ (V_0), with electrons leaving the cathode at zero speed.

RESULTS

$$V(y)=V_0\left(\frac{y}{d}\right)^{4/3},\qquad J=\frac{4\epsilon_0}{9d^2}\sqrt{\frac{2e}{m}}\,V_0^{3/2}\quad\text{(Child–Langmuir law)}$$

Interpretation: the diode is non-linear ($J\propto V_0^{3/2}$), unlike Ohm's law. (Full derivation in Q16.)

(c) Spherical electron cloud of uniform density $-\rho_0$, radius b (Cheng Ex. 4-3): inside, Poisson gives $\mathbf{E}=-\mathbf{a}_R\dfrac{\rho_0R}{3\epsilon_0}$. Outside, Laplace gives $\mathbf{E}=-\mathbf{a}_R\dfrac{\rho_0b^3}{3\epsilon_0R^2}$. The constants come from "E finite at R = 0" and "E continuous at R = b".

8 Part B4: Separation of variables (when V depends on two coordinates)

When V depends on two coordinates (say x and y), guess $V(x,y) = X(x)\,Y(y)$. Substituting into $\nabla^2 V = 0$ and dividing by XY:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} = 0$$

The first term depends only on x and the second only on y, and they sum to zero for *all* x and y. So each must be a constant: $\frac{X''}{X} = -k^2,\ \frac{Y''}{Y} = +k^2$ (or the opposite sign). k is the **separation constant**.

k^2	$X(x)$	Shape and when you need it
$X'' = 0$ ($k = 0$)	$A_0 x + B_0$	Linear
$X'' = -k^2 X$	$A\sin kx + B\cos kx$	Oscillating. Use in the direction where **V = 0 on two parallel boundaries** (so you can fit zeros).
$X'' = +k^2 X$	$Ae^{kx} + Be^{-kx}$ or $A\sinh kx + B\cosh kx$	Growing or decaying. Use in the direction going to infinity (e^{-kx}) or a finite box ($\sinh$).

Sum of the separation constants: $k_x^2 + k_y^2 + k_z^2 = 0$ (in 3-D), so at least one must be imaginary. In other words, at least one direction oscillates and at least one grows or decays.

Classic problem: the grounded semi-infinite slot

Two grounded half-planes at $y = 0$ and $y = b$ extend from $x = 0$ to $x = \infty$. The end at $x = 0$ is an insulated strip held at V_0. Find V in the slot. (Fully worked in Q21.)

$$V(x,y) = \frac{4V_0}{\pi}\sum_{n=1,3,5,\ldots}\frac{1}{n}\,e^{-n\pi x/b}\sin\frac{n\pi y}{b}$$

Advanced results in cylindrical and spherical coordinates (recognise them)

- Cylindrical, $V(r,\phi)$: separated solutions are $r^{\pm n}(\sin n\phi, \cos n\phi)$. A conducting cylinder of radius a placed in a uniform field $\mathbf{E}_0 = \mathbf{a}_x E_0$ gives $V = -E_0\left(r - \frac{a^2}{r}\right)\cos\phi$, with $\rho_s = 2\epsilon_0 E_0 \cos\phi$.
- Spherical, $V(R,\theta)$: solutions are $R^n P_n(\cos\theta)$ and $R^{-(n+1)}P_n(\cos\theta)$, where P_n are the Legendre polynomials $P_0 = 1,\ P_1 = \cos\theta,\ P_2 = \frac{1}{2}(3\cos^2\theta - 1)$. A conducting sphere of radius b in a uniform field $\mathbf{a}_z E_0$ gives $V = -E_0\left(1 - \frac{b^3}{R^3}\right)R\cos\theta$, with $\rho_s = 3\epsilon_0 E_0 \cos\theta$.

9 Formula sheet

ENERGY

$$W_e = \frac{1}{2}\sum Q_k V_k = \frac{1}{4\pi\epsilon_0}\sum_{\text{pairs}} \frac{Q_i Q_j}{R_{ij}}$$

$$W_e = \frac{1}{2}\int \rho V \, dv$$

$$W_e = \frac{1}{2}\int \mathbf{D}\cdot\mathbf{E}\, dv = \frac{1}{2}\int \epsilon E^2 dv = \frac{1}{2}\int \frac{D^2}{\epsilon} dv$$

$$w_e = \frac{1}{2}\mathbf{D}\cdot\mathbf{E} = \frac{1}{2}\epsilon E^2 = \frac{D^2}{2\epsilon} \text{ J/m}^3$$

$$W_e = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$$

$$C = \frac{2W_e}{V^2} = \frac{Q^2}{2W_e}$$

$$1\text{ eV} = 1.6\times10^{-19}\text{ J}$$

FORCES

$\mathbf{F}_Q = -\nabla W_e$ (Q constant)

$\mathbf{F}_V = +\nabla W_e$ (V constant)

$T_z = \mp \partial W_e/\partial\phi$

Pressure on a conductor: $p = \dfrac{\rho_s^2}{2\epsilon} = \dfrac{1}{2}\epsilon E^2$ N/m²

Plate force: $F = \dfrac{Q^2}{2\epsilon S} = \dfrac{\epsilon S V^2}{2d^2}$ (attractive)

POISSON / LAPLACE

$$\nabla\cdot(\epsilon\nabla V) = -\rho$$

$\nabla^2 V = -\rho/\epsilon$ (Poisson)

$\nabla^2 V = 0$ (Laplace)

$\mathbf{E} = -\nabla V, \quad \rho_s = \epsilon E_n$

Uniqueness: equation + boundary conditions give exactly one solution.

1-D SOLUTIONS

$V = C_1 y + C_2$
$V = C_1 \ln r + C_2$
$V = C_1\phi + C_2$
$V = -C_1/R + C_2$
$V = C_1 \ln\tan(\theta/2) + C_2$

Diode: $V = V_0(y/d)^{4/3}$, $J = \dfrac{4\epsilon_0}{9d^2}\sqrt{\dfrac{2e}{m}}V_0^{3/2}$

LAPLACIANS

$\nabla^2 V = V_{xx} + V_{yy} + V_{zz}$
$\nabla^2 V = \frac{1}{r}\partial_r(rV_r) + \frac{1}{r^2}V_{\phi\phi} + V_{zz}$
$\nabla^2 V = \frac{1}{R^2}\partial_R(R^2 V_R) + \frac{1}{R^2\sin\theta}\partial_\theta(\sin\theta V_\theta) + \frac{1}{R^2\sin^2\theta}V_{\phi\phi}$

10 Practice questions with step-by-step solutions

Try each question on paper first (cover the solution with a sheet), then read the solution. Unless stated otherwise, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m and $1/(4\pi\epsilon_0) \approx 9 \times 10^9$.

Set A: Energy of charge systems

Q1. VERY LIKELY

Three point charges $Q_1 = 1\ \mu\text{C},\ Q_2 = 2\ \mu\text{C},\ Q_3 = -3\ \mu\text{C}$ sit at the corners of an equilateral triangle of side 10 cm in free space. Find the stored electrostatic energy.

STEP-BY-STEP SOLUTION

1. **Choose the formula.** Discrete point charges, so use the pair-sum form: $W_e = \dfrac{1}{4\pi\epsilon_0}\left(\dfrac{Q_1Q_2}{R_{12}} + \dfrac{Q_1Q_3}{R_{13}} + \dfrac{Q_2Q_3}{R_{23}}\right)$.
2. **All distances are equal** ($R = 0.1$ m), so factor out: $W_e = \dfrac{1}{4\pi\epsilon_0 R}(Q_1Q_2 + Q_1Q_3 + Q_2Q_3)$.
3. **Products, keeping signs** (in μC², with 1 μC² = 10^{-12} C²): $1\cdot 2 = 2,\ 1\cdot(-3) = -3,\ 2\cdot(-3) = -6$. Sum = -7×10^{-12} C².
4. **Substitute:** $W_e = \dfrac{9 \times 10^9 \times (-7 \times 10^{-12})}{0.1} = -0.63$ J.

> **Answer: $W_e = -0.63$ J.**

Interpretation: the energy is negative because attraction dominates. The field did net work as the charges were brought together, so the system is bound: you would need 0.63 J to pull them apart to infinity.

Q2. VERY LIKELY

Four equal charges Q are placed at the corners of a square of side a. Find the energy required to assemble them.

STEP-BY-STEP SOLUTION

1. **Count the pairs.** 4 charges make $\binom{4}{2} = 6$ pairs: 4 along the sides (distance a) and 2 along the diagonals (distance $a\sqrt{2}$).
2. $W_e = \dfrac{Q^2}{4\pi\epsilon_0}\left(\dfrac{4}{a} + \dfrac{2}{a\sqrt{2}}\right) = \dfrac{Q^2}{4\pi\epsilon_0 a}\left(4 + \sqrt{2}\right)$.
3. Numerically: $4 + \sqrt{2} = 5.414$.

Answer: $W_e = \dfrac{5.414\,Q^2}{4\pi\epsilon_0 a} \approx \dfrac{0.431\,Q^2}{\epsilon_0 a}$.

Cross-check with $\frac{1}{2}\sum Q_k V_k$: each charge sees potential $V = \dfrac{Q}{4\pi\epsilon_0}\left(\dfrac{2}{a} + \dfrac{1}{a\sqrt{2}}\right)$. Then $W = \frac{1}{2}\cdot 4 \cdot Q \cdot V = \dfrac{Q^2}{4\pi\epsilon_0 a}(4+\sqrt{2})$. The same, as it should be.

Q3. DIRECT

In a hydrogen atom the electron is 0.53 Å (0.53×10^{-10} m) from the proton. Find the electrostatic potential energy in joules and in eV.

STEP-BY-STEP SOLUTION

1. Two charges: $W_e = \dfrac{Q_1 Q_2}{4\pi\epsilon_0 R}$, with $Q_1 = +e$ and $Q_2 = -e$, $e = 1.6 \times 10^{-19}$ C.
2. $W_e = \dfrac{9 \times 10^9 \times (-(1.6\times10^{-19})^2)}{0.53 \times 10^{-10}} = \dfrac{-9 \times 10^9 \times 2.56 \times 10^{-38}}{5.3 \times 10^{-11}} = -4.35 \times 10^{-18}$ J.
3. To eV: divide by 1.6×10^{-19}: $-4.35 \times 10^{-18}/1.6 \times 10^{-19} = -27.2$ eV.

Answer: $W_e \approx -4.35 \times 10^{-18}$ J $= -27.2$ eV.

Q4. VERY LIKELY

Find the total energy stored in a sphere of radius b carrying a uniform volume charge density ρ (total charge Q), (a) using $W = \frac{1}{2}\int \rho V\, dv$ and (b) using $W = \frac{1}{2}\int \epsilon_0 E^2 dv$. (c) What fraction of the energy is stored inside the sphere?

STEP-BY-STEP SOLUTION

First, find E by Gauss's law (we'll need it):

- Inside ($R < b$): enclosed charge $= \rho \cdot \frac{4}{3}\pi R^3$. Then $E \cdot 4\pi R^2 = \dfrac{\rho \frac{4}{3}\pi R^3}{\epsilon_0} \Rightarrow E_1 = \dfrac{\rho R}{3\epsilon_0}$.
- Outside ($R > b$): $E_2 = \dfrac{\rho b^3}{3\epsilon_0 R^2} = \dfrac{Q}{4\pi\epsilon_0 R^2}$.

(a) Using $\frac{1}{2}\int \rho V\, dv$. We need V inside.

1. $V(R) = -\int_\infty^b E_2\, dR - \int_b^R E_1\, dR = \dfrac{\rho b^2}{3\epsilon_0} + \dfrac{\rho}{6\epsilon_0}(b^2 - R^2) = \dfrac{\rho}{3\epsilon_0}\left(\dfrac{3}{2}b^2 - \dfrac{R^2}{2}\right)$.
2. $W_e = \dfrac{\rho}{2}\int_0^b \dfrac{\rho}{3\epsilon_0}\left(\dfrac{3}{2}b^2 - \dfrac{R^2}{2}\right) 4\pi R^2 dR$.
3. $\int_0^b \left(\frac{3}{2}b^2R^2 - \frac{1}{2}R^4\right)dR = \frac{1}{2}b^5 - \frac{1}{10}b^5 = \frac{2}{5}b^5$.

4. $W_e = \dfrac{\rho^2}{6\epsilon_0}\cdot 4\pi\cdot\dfrac{2}{5}b^5 = \dfrac{4\pi\rho^2 b^5}{15\epsilon_0}$.

5. With $\rho = \dfrac{3Q}{4\pi b^3}$: $W_e = \dfrac{4\pi b^5}{15\epsilon_0}\cdot\dfrac{9Q^2}{16\pi^2 b^6} = \dfrac{3Q^2}{20\pi\epsilon_0 b}$.

(b) Using $\frac{1}{2}\int \epsilon_0 E^2\,dv$, split into the two regions:

1. Inside: $W_{in} = \frac{1}{2}\epsilon_0\int_0^b\left(\dfrac{QR}{4\pi\epsilon_0 b^3}\right)^2 4\pi R^2 dR = \dfrac{Q^2}{8\pi\epsilon_0 b^6}\int_0^b R^4 dR = \dfrac{Q^2}{40\pi\epsilon_0 b}$.

2. Outside: $W_{out} = \frac{1}{2}\epsilon_0\int_b^\infty\left(\dfrac{Q}{4\pi\epsilon_0 R^2}\right)^2 4\pi R^2 dR = \dfrac{Q^2}{8\pi\epsilon_0}\int_b^\infty\dfrac{dR}{R^2} = \dfrac{Q^2}{8\pi\epsilon_0 b}$.

3. Total: $\dfrac{Q^2}{8\pi\epsilon_0 b}\left(1+\dfrac{1}{5}\right) = \dfrac{6Q^2}{40\pi\epsilon_0 b} = \dfrac{3Q^2}{20\pi\epsilon_0 b}$. This matches (a).

(c) Fraction inside $= \dfrac{1/40}{6/40} = \dfrac{1}{6}$.

Answer: $W_e = \dfrac{3Q^2}{20\pi\epsilon_0 b} = \dfrac{4\pi\rho^2 b^5}{15\epsilon_0}$. One sixth of it is stored inside the sphere and five sixths outside.

Q5. VERY LIKELY

A metal sphere of radius $a = 10$ cm in air carries charge $Q = 1\ \mu$C. Find (a) the energy stored, (b) the energy density at its surface.

STEP-BY-STEP SOLUTION

1. Outside, $E = \dfrac{Q}{4\pi\epsilon_0 R^2}$. Inside a conductor E = 0, so there is no energy inside.

2. $W_e = \int_a^\infty \frac{1}{2}\epsilon_0 E^2\,4\pi R^2 dR = \dfrac{Q^2}{8\pi\epsilon_0 a}$. Check: $C = 4\pi\epsilon_0 a$ and $Q^2/2C = Q^2/(8\pi\epsilon_0 a)$. It agrees.

3. Numbers: $\dfrac{Q^2}{8\pi\epsilon_0 a} = \dfrac{1}{2}\cdot\dfrac{Q^2}{4\pi\epsilon_0 a} = \dfrac{1}{2}\cdot\dfrac{9\times10^9\times10^{-12}}{0.1} = \dfrac{1}{2}\times 0.09 = 0.045$ J.

4. At the surface: $E = \dfrac{9\times10^9\times10^{-6}}{0.01} = 9\times10^5$ V/m. Then
$w_e = \frac{1}{2}\epsilon_0 E^2 = 0.5\times8.854\times10^{-12}\times8.1\times10^{11} = 3.59$ J/m³.

Answer: (a) 0.045 J = 45 mJ, (b) 3.59 J/m³.

Set B: Energy density and capacitors

Q6. VERY LIKELY

In free space, $V = 3x^2 + 2y$ (V). Find the electrostatic energy stored in the cube $0 \le x, y, z \le 1$ m.

STEP-BY-STEP SOLUTION

1. $\mathbf{E} = -\nabla V = -\left(\mathbf{a}_x \dfrac{\partial V}{\partial x} + \mathbf{a}_y \dfrac{\partial V}{\partial y}\right) = -6x\,\mathbf{a}_x - 2\,\mathbf{a}_y$ V/m.
2. $E^2 = 36x^2 + 4$. Then $w_e = \frac{1}{2}\epsilon_0(36x^2 + 4)$.
3. $W_e = \int_0^1\int_0^1\int_0^1 \frac{1}{2}\epsilon_0(36x^2+4)\,dx\,dy\,dz = \frac{1}{2}\epsilon_0\left[12x^3 + 4x\right]_0^1 \cdot 1 \cdot 1 = \frac{1}{2}\epsilon_0(16) = 8\epsilon_0$.
4. $8 \times 8.854 \times 10^{-12} = 7.08 \times 10^{-11}$ J.

Answer: $W_e = 8\epsilon_0 \approx 70.8$ pJ.

Side note: $\nabla^2 V = 6 \neq 0$, so this V is *not* a Laplace solution. The region holds charge $\rho = -\epsilon_0 \nabla^2 V = -6\epsilon_0$. The energy formula still works.

Q7. VERY LIKELY

A parallel-plate capacitor has plate area 100 cm², separation 2 mm, a dielectric with $\epsilon_r = 4$, and 100 V applied. Find C, Q, E, the energy density, and the total energy (compute it two ways).

STEP-BY-STEP SOLUTION

1. **Units first:** $S = 100\ \text{cm}^2 = 0.01\ \text{m}^2$, $d = 0.002$ m, $\epsilon = 4\epsilon_0 = 3.5416 \times 10^{-11}$ F/m.
2. $C = \dfrac{\epsilon S}{d} = \dfrac{3.5416 \times 10^{-11} \times 0.01}{0.002} = 1.771 \times 10^{-10}$ F = 177.1 pF.
3. $Q = CV = 1.771 \times 10^{-8}$ C = 17.7 nC.
4. $E = V/d = 100/0.002 = 5 \times 10^4$ V/m.
5. $w_e = \frac{1}{2}\epsilon E^2 = 0.5 \times 3.5416 \times 10^{-11} \times 2.5 \times 10^9 = 0.0443$ J/m³.
6. **Way 1:** $W = w_e \times \text{volume} = 0.0443 \times (0.01 \times 0.002) = 8.85 \times 10^{-7}$ J.
   Way 2: $W = \frac{1}{2}CV^2 = 0.5 \times 1.771 \times 10^{-10} \times 10^4 = 8.85 \times 10^{-7}$ J. Both agree.

Answer: C = 177 pF, Q = 17.7 nC, E = 50 kV/m, w_e = 0.0443 J/m³, W = 0.885 µJ.

Q8. CONCEPTUAL, VERY LIKELY

A charged air parallel-plate capacitor has its plate separation doubled. Find the factor by which the stored energy changes if (a) the battery was disconnected first, (b) the battery remains connected. Where does the energy go or come from?

STEP-BY-STEP SOLUTION

1. $C = \epsilon S/d$, so doubling d gives $C' = C/2$.
2. **(a) Disconnected, so Q is fixed.** Use the form that contains Q: $W = Q^2/2C$. Then $W' = Q^2/(2 \cdot C/2) = 2W$. **Energy doubles.** It came from the mechanical work you did pulling the attracting plates apart. (E = σ/ε stays the same, and the volume of field doubles.)
3. **(b) Connected, so V is fixed.** Use $W = \frac{1}{2}CV^2$. Then $W' = \frac{1}{2}(C/2)V^2 = W/2$. **Energy halves.** Q drops by $Q/2$, so charge $Q/2$ is pushed back into the battery. The battery absorbs $V \cdot Q/2 = W$. That equals the lost field energy ($W/2$) plus your mechanical work ($W/2$).

Answer: (a) ×2, (b) ×½.

RULE

Always write W using the quantity that is held constant.

Q9. VERY LIKELY

Using the energy method, find the capacitance per unit length of a coaxial cable with inner radius a, outer radius b, and dielectric ϵ. Evaluate for $a = 1$ mm, $b = 4$ mm, $\epsilon_r = 2.25$, and find the energy stored per metre at 100 V.

STEP-BY-STEP SOLUTION

1. **Assume charge $+Q$** on the inner conductor (length L). Gauss with a cylinder of radius r: $D \cdot 2\pi r L = Q \Rightarrow E = \dfrac{Q}{2\pi\epsilon L r}$ for $a < r < b$.
2. **Energy:** volume element $dv = 2\pi r L\, dr$. $W_e = \frac{1}{2}\int_a^b \epsilon \left(\dfrac{Q}{2\pi\epsilon L r}\right)^2 2\pi r L\, dr = \dfrac{Q^2}{4\pi\epsilon L}\int_a^b \dfrac{dr}{r} = \dfrac{Q^2}{4\pi\epsilon L}\ln\dfrac{b}{a}$.
3. **Equate to $Q^2/2C$:** $\dfrac{Q^2}{2C} = \dfrac{Q^2}{4\pi\epsilon L}\ln\dfrac{b}{a} \Rightarrow C = \dfrac{2\pi\epsilon L}{\ln(b/a)}$.
4. **Numbers** (L = 1 m): $2\pi \times 2.25 \times 8.854 \times 10^{-12} = 1.2517 \times 10^{-10}$; $\ln 4 = 1.3863$. So $C = 9.03 \times 10^{-11}$ F/m = 90.3 pF/m.
5. $W = \frac{1}{2}CV^2 = 0.5 \times 9.03 \times 10^{-11} \times 10^4 = 4.51 \times 10^{-7}$ J/m.

Answer: $C = 2\pi\epsilon / \ln(b/a)$ per metre = 90.3 pF/m, W = 0.451 μJ/m.

Q10. LIKELY

A parallel-plate capacitor (area 0.01 m²) has two dielectric layers in series: layer 1 is 1 mm thick with $\epsilon_{r1} = 2$, and layer 2 is 1 mm thick with $\epsilon_{r2} = 4$. 300 V is applied. Find D, E and w_e in each layer, and the total energy. Which layer stores more energy?

STEP-BY-STEP SOLUTION

1. **Key physics:** at the interface, the normal D is continuous (no free surface charge). So D is the *same* in both layers, but $E_i = D/\epsilon_i$ differs.

2. Voltage adds: $V = E_1 d_1 + E_2 d_2 = D\left(\frac{d_1}{\epsilon_1} + \frac{d_2}{\epsilon_2}\right)$.
   $\frac{d_1}{\epsilon_1} + \frac{d_2}{\epsilon_2} = \frac{10^{-3}}{\epsilon_0}\left(\frac{1}{2} + \frac{1}{4}\right) = \frac{0.75 \times 10^{-3}}{\epsilon_0}$.
   $D = \frac{300\,\epsilon_0}{0.75 \times 10^{-3}} = 4 \times 10^5 \epsilon_0 = 3.54 \times 10^{-6}$ C/m².

3. $E_1 = D/(2\epsilon_0) = 2 \times 10^5$ V/m and $E_2 = D/(4\epsilon_0) = 1 \times 10^5$ V/m. Check: $200 + 100 = 300$ V. ✓

4. $w_{e1} = \frac{1}{2}\epsilon_1 E_1^2 = \frac{1}{2}(2\epsilon_0)(4 \times 10^{10}) = 4 \times 10^{10}\epsilon_0 = 0.354$ J/m³.
   $w_{e2} = \frac{1}{2}(4\epsilon_0)(10^{10}) = 2 \times 10^{10}\epsilon_0 = 0.177$ J/m³.

5. $W = (0.354 + 0.177) \times (0.01 \times 10^{-3}) = 5.31 \times 10^{-6}$ J.
   Check: $C = \frac{S}{d_1/\epsilon_1 + d_2/\epsilon_2} = \frac{0.01\,\epsilon_0}{0.75 \times 10^{-3}} = 1.18 \times 10^{-10}$ F, and $\frac{1}{2}CV^2 = 5.31 \times 10^{-6}$ J. ✓

Answer: the lower-permittivity layer ($\epsilon_r = 2$) has twice the field and stores twice the energy.

Interpretation: with D fixed, $w_e = D^2/2\epsilon$. A smaller ε means more energy. This is why an air gap in series with insulation is the most stressed part and breaks down first.

Set C: Forces from energy

Q11. VERY LIKELY

Find the force between the plates of a parallel-plate capacitor (area S, separation x, permittivity ε) using the principle of virtual displacement, (a) with constant charge Q and (b) with constant voltage V. Evaluate for the capacitor of Q7.

STEP-BY-STEP SOLUTION

1. **(a) Fixed Q.** Write W in terms of Q and x: $W_e = \frac{Q^2}{2C} = \frac{Q^2 x}{2\epsilon S}$.

2. $F_x = -\dfrac{\partial W_e}{\partial x} = -\dfrac{Q^2}{2\epsilon S}$. The minus sign means the force acts in the direction of decreasing x, so the plates **attract**.

3. **(b) Fixed V.** $W_e = \frac{1}{2}CV^2 = \dfrac{\epsilon S V^2}{2x}$. Then $F_x = +\dfrac{\partial W_e}{\partial x} = -\dfrac{\epsilon S V^2}{2x^2}$. Also attractive.

4. **Same?** Substitute $Q = \epsilon S V/x$ into (a): $\dfrac{\epsilon^2 S^2 V^2/x^2}{2\epsilon S} = \dfrac{\epsilon S V^2}{2x^2}$. ✓ The two cases give the same force.

5. **Numbers (Q7):** $|F| = \frac{1}{2}\epsilon E^2 S = 0.0443 \times 0.01 = 4.43 \times 10^{-4}$ N.

Answer: $|F| = \dfrac{Q^2}{2\epsilon S} = \dfrac{\epsilon S V^2}{2x^2}$, attractive; 0.443 mN for Q7.

Interpretation: force per area = $\frac{1}{2}\epsilon E^2$ = energy density. Field lines act like stretched rubber bands pulling the plates together.

Q12. LIKELY

The surface charge density on a conductor in air is $\rho_s = 10\ \mu$C/m². Find the electrostatic pressure on the surface and its direction.

STEP-BY-STEP SOLUTION

1. At the conductor surface $E = \rho_s/\epsilon_0$, so $p = \frac{1}{2}\epsilon_0 E^2 = \dfrac{\rho_s^2}{2\epsilon_0}$.

2. $p = \dfrac{(10^{-5})^2}{2 \times 8.854 \times 10^{-12}} = \dfrac{10^{-10}}{1.7708 \times 10^{-11}} = 5.65$ N/m².

Answer: 5.65 N/m², directed outward (normal to the surface, away from the conductor), regardless of the sign of the charge.

Set D: Laplace's equation, checking and solving

Q13. VERY LIKELY (DIRECT)

Which of the following satisfy Laplace's equation? For any that don't, find the charge density in free space that produces it.
(i) $V = x^2 - y^2$ (ii) $V = x^2 + y^2$ (iii) $V = e^{-x}\sin y$ (iv) $V = \ln r$ (cylindrical) (v) $V = r\cos\phi$ (vi) $V = \dfrac{\cos\theta}{R^2}$ (vii) $V = R^2$ (spherical)

STEP-BY-STEP SOLUTION

Method: compute $\nabla^2 V$ in the matching coordinate system. If it equals 0, V is harmonic. Otherwise $\rho = -\epsilon_0 \nabla^2 V$.

i. $V_{xx} = 2,\ V_{yy} = -2$. Sum = 0. ✓ **Satisfies.**

ii. $2 + 2 = 4 \neq 0$. × $\rho = -4\epsilon_0 = -3.54 \times 10^{-11}$ C/m³.

iii. $V_{xx} = e^{-x}\sin y,\ V_{yy} = -e^{-x}\sin y$. Sum = 0. ✓

iv. $\frac{1}{r}\frac{d}{dr}\left(r \cdot \frac{1}{r}\right) = \frac{1}{r}\frac{d}{dr}(1) = 0$. ✓ (This is the coax solution.)

v. $\frac{1}{r}\partial_r(r\cos\phi) + \frac{1}{r^2}(-r\cos\phi) = \frac{\cos\phi}{r} - \frac{\cos\phi}{r} = 0$. ✓ (It's just $V = x$.)

vi. R-part: $\frac{1}{R^2}\partial_R\left(R^2 \cdot (-2)\frac{\cos\theta}{R^3}\right) = \frac{1}{R^2}\partial_R\left(-\frac{2\cos\theta}{R}\right) = \frac{2\cos\theta}{R^4}$.
θ-part: $\frac{1}{R^2\sin\theta}\partial_\theta\left(\sin\theta \cdot \frac{-\sin\theta}{R^2}\right) = \frac{-2\sin\theta\cos\theta}{R^4\sin\theta} = -\frac{2\cos\theta}{R^4}$.
Sum = 0. ✓ (This is the potential of a dipole.)

vii. $\frac{1}{R^2}\partial_R(R^2 \cdot 2R) = \frac{6R^2}{R^2} = 6 \neq 0$. × $\rho = -6\epsilon_0$.

Answer: (i), (iii), (iv), (v), (vi) are harmonic. For (ii) $\rho = -4\epsilon_0$ and for (vii) $\rho = -6\epsilon_0$.

Q14. VERY LIKELY (CHENG EX 4-1)

Parallel plates: the lower plate ($y = 0$) is grounded, the upper plate ($y = d$) is at V_0, and fringing is neglected. Find V(y), E, and the surface charge density on each plate. Then take $d = 5$ mm, $V_0 = 200$ V, $\epsilon_r = 2.5$.

STEP-BY-STEP SOLUTION

1. **Which equation?** No charge between the plates, so use Laplace. No fringing means V depends only on y: $\frac{d^2V}{dy^2} = 0$.

2. **Integrate twice:** $\frac{dV}{dy} = C_1 \Rightarrow V = C_1 y + C_2$.

3. **Boundary conditions:** $V(0) = 0 \Rightarrow C_2 = 0$. $V(d) = V_0 \Rightarrow C_1 = V_0/d$. So $V = \frac{V_0}{d}y$.

4. $\mathbf{E} = -\nabla V = -\mathbf{a}_y\frac{V_0}{d}$. It points down, from high V to low V. ✓

5. **Charges.** $\mathbf{D} = -\mathbf{a}_y\epsilon V_0/d$.
Lower plate: normal pointing into the gap is $+\mathbf{a}_y$, so $\rho_{s,\text{lower}} = \mathbf{a}_y \cdot \mathbf{D} = -\epsilon V_0/d$.
Upper plate: normal into the gap is $-\mathbf{a}_y$, so $\rho_{s,\text{upper}} = +\epsilon V_0/d$.

6. **Numbers:** $E = 200/0.005 = 4 \times 10^4$ V/m.
$\rho_s = \epsilon E = 2.5 \times 8.854 \times 10^{-12} \times 4 \times 10^4 = 8.85 \times 10^{-7}$ C/m².

Answer: $V = 40{,}000\,y$ V, $\mathbf{E} = -4 \times 10^4\mathbf{a}_y$ V/m, $\rho_s = \pm 0.885\ \mu$C/m² (+ on the top plate, − on the bottom).

Q15. LIKELY (POISSON)

The region $0 < y < d$ between two grounded plates holds a uniform charge density ρ_0. Find V(y), E(y), the maximum potential, and its location.

STEP-BY-STEP SOLUTION

1. Charge is present, so use **Poisson**: $\frac{d^2V}{dy^2} = -\frac{\rho_0}{\epsilon}$.
2. Integrate once: $\frac{dV}{dy} = -\frac{\rho_0}{\epsilon}y + C_1$. Integrate again: $V = -\frac{\rho_0}{2\epsilon}y^2 + C_1 y + C_2$.
3. $V(0) = 0 \Rightarrow C_2 = 0$. $V(d) = 0 \Rightarrow -\frac{\rho_0 d^2}{2\epsilon} + C_1 d = 0 \Rightarrow C_1 = \frac{\rho_0 d}{2\epsilon}$.
4. $V(y) = \frac{\rho_0}{2\epsilon}(yd - y^2) = \frac{\rho_0}{2\epsilon}y(d-y)$.
5. $\mathbf{E} = -\frac{dV}{dy}\mathbf{a}_y = -\frac{\rho_0}{2\epsilon}(d-2y)\,\mathbf{a}_y = \frac{\rho_0}{\epsilon}\left(y - \frac{d}{2}\right)\mathbf{a}_y$. E is zero at the centre and points outward toward both plates (for $\rho_0 > 0$).
6. Maximum: $dV/dy = 0 \Rightarrow y = d/2$. $V_{max} = \frac{\rho_0}{2\epsilon}\cdot\frac{d}{2}\cdot\frac{d}{2} = \frac{\rho_0 d^2}{8\epsilon}$.

Answer: $V = \frac{\rho_0}{2\epsilon}y(d-y)$, with $V_{max} = \frac{\rho_0 d^2}{8\epsilon}$ at the midplane.

Note that this V has an interior maximum. That is allowed only because charge is present: with Poisson, unlike Laplace, interior maxima can occur.

Q16. LIKELY (CHENG EX 4-2)

Vacuum diode: cathode at $y = 0$ (V = 0), anode at $y = d$ ($V = V_0$). Electrons leave the cathode with zero velocity and the field at the cathode is zero (space-charge limited). Derive V(y) and the current density J.

STEP-BY-STEP SOLUTION

1. **The electrons are a volume charge**, so use Poisson: $\frac{d^2V}{dy^2} = -\frac{\rho}{\epsilon_0}$, where ρ is negative.
2. **Electron speed** from energy conservation: $\frac{1}{2}mu^2 = eV \Rightarrow u = \sqrt{2eV/m}$.
3. **Current is the same everywhere** in the steady state: $J = |\rho|u \Rightarrow \rho = -\frac{J}{u} = -\frac{J}{\sqrt{2e/m}}V^{-1/2}$.
4. Poisson becomes $\frac{d^2V}{dy^2} = \frac{J}{\epsilon_0\sqrt{2e/m}}V^{-1/2} \equiv K V^{-1/2}$.
5. **Trick:** multiply both sides by $2\frac{dV}{dy}$. The left side becomes $\frac{d}{dy}\left(\frac{dV}{dy}\right)^2$ and the right side becomes $4K\frac{d}{dy}\left(V^{1/2}\right)$.

6. Integrate. At $y = 0$, both V and dV/dy are 0, so the constant is 0:
$\left(\frac{dV}{dy}\right)^2 = 4KV^{1/2} \Rightarrow \frac{dV}{dy} = 2\sqrt{K}\,V^{1/4}$.

7. Separate: $V^{-1/4}dV = 2\sqrt{K}\,dy \Rightarrow \frac{4}{3}V^{3/4} = 2\sqrt{K}\,y$ (the constant is 0 since V(0) = 0).

8. At $y = d$: $\frac{4}{3}V_0^{3/4} = 2\sqrt{K}d$. Divide the two: $\left(\frac{V}{V_0}\right)^{3/4} = \frac{y}{d} \Rightarrow V = V_0\left(\frac{y}{d}\right)^{4/3}$.

9. From step 8, $\sqrt{K} = \frac{2V_0^{3/4}}{3d} \Rightarrow K = \frac{4V_0^{3/2}}{9d^2}$. Since $K = \frac{J}{\epsilon_0\sqrt{2e/m}}$, $J = \frac{4\epsilon_0}{9d^2}\sqrt{\frac{2e}{m}}\,V_0^{3/2}$.

Answer: $V = V_0(y/d)^{4/3}$ and $J = \frac{4\epsilon_0}{9d^2}\sqrt{\frac{2e}{m}}V_0^{3/2}$ (the Child–Langmuir three-halves power law).

Q17. VERY LIKELY

Coaxial cable: inner conductor radius $a = 1$ mm at $V_0 = 100$ V, outer conductor radius $b = 5$ mm grounded. Solve Laplace's equation for V(r), find E, the potential at $r = 2$ mm, the surface charge on the inner conductor, and C per unit length ($\epsilon_r = 1$).

STEP-BY-STEP SOLUTION

1. By symmetry V depends only on r: $\frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0$.

2. Multiply by r and integrate: $r\frac{dV}{dr} = C_1 \Rightarrow V = C_1 \ln r + C_2$.

3. $V(b) = 0 \Rightarrow C_2 = -C_1 \ln b$, so $V = C_1 \ln(r/b) = -C_1 \ln(b/r)$.
$V(a) = V_0 \Rightarrow -C_1 \ln(b/a) = V_0 \Rightarrow C_1 = -\frac{V_0}{\ln(b/a)}$.
Therefore $V(r) = V_0\frac{\ln(b/r)}{\ln(b/a)}$.

4. $\mathbf{E} = -\mathbf{a}_r\frac{dV}{dr} = \mathbf{a}_r\frac{V_0}{r\ln(b/a)}$, pointing outward from the high-V inner conductor. ✓

5. At r = 2 mm: $V = 100 \times \frac{\ln 2.5}{\ln 5} = 100 \times \frac{0.9163}{1.6094} = 56.9$ V. *Not* 75 V: the potential falls off logarithmically, not linearly.

6. $\rho_s(a) = \epsilon_0 E(a) = \frac{\epsilon_0 V_0}{a\ln(b/a)} = \frac{8.854\times10^{-12}\times 100}{10^{-3}\times 1.6094} = 5.50\times10^{-7}$ C/m².

7. Charge per metre: $Q' = 2\pi a\rho_s = \frac{2\pi\epsilon_0 V_0}{\ln(b/a)}$. So $C' = \frac{Q'}{V_0} = \frac{2\pi\epsilon_0}{\ln 5} = \frac{5.563\times10^{-11}}{1.6094} = 34.6$ pF/m.

Answer: $V = V_0\ln(b/r)/\ln(b/a)$, E = 62.1/r V/m (r in m), V(2 mm) = 56.9 V, ρ_s = 0.55 μC/m², C = 34.6 pF/m.

Q18. VERY LIKELY

Concentric spheres: the inner sphere ($R = a = 5$ cm) is at $V_0 = 100$ V and the outer ($R = b = 10$ cm) is grounded. Solve Laplace's equation for V(R), find V at R = 7.5 cm, E at the inner surface, and the capacitance (air).

STEP-BY-STEP SOLUTION

1. V depends only on R: $\frac{1}{R^2}\frac{d}{dR}\left(R^2\frac{dV}{dR}\right) = 0 \Rightarrow R^2\frac{dV}{dR} = C_1 \Rightarrow V = -\frac{C_1}{R} + C_2$.

2. Write it as $V = \frac{A}{R} + B$. $V(b) = 0 \Rightarrow B = -A/b$, so $V = A\left(\frac{1}{R} - \frac{1}{b}\right)$.
   $V(a) = V_0 \Rightarrow A = \frac{V_0}{\frac{1}{a} - \frac{1}{b}}$.

3. $V(R) = V_0\frac{\frac{1}{R} - \frac{1}{b}}{\frac{1}{a} - \frac{1}{b}}$. With a = 0.05, b = 0.1: $\frac{1}{a} - \frac{1}{b} = 20 - 10 = 10$ m^{-1}.

4. At R = 0.075: $\frac{1}{R} = 13.33$. So $V = 100 \times \frac{13.33 - 10}{10} = 33.3$ V.

5. $E_R = -\frac{dV}{dR} = \frac{A}{R^2} = \frac{V_0}{R^2(1/a - 1/b)}$. At R = a: $\frac{100}{0.0025 \times 10} = 4000$ V/m.

6. $Q = 4\pi a^2 \epsilon_0 E(a) = 4\pi\epsilon_0 A = \frac{4\pi\epsilon_0 V_0}{1/a - 1/b}$. So $C = \frac{4\pi\epsilon_0}{1/a - 1/b} = \frac{1.1127 \times 10^{-10}}{10} = 11.1$ pF.

Answer: V(7.5 cm) = 33.3 V, E(a) = 4 kV/m, C = 11.1 pF.

Q19. LIKELY

Two infinite conducting half-planes meet along the z-axis at an angle $\alpha = 30^\circ$, separated by a thin insulating gap at the edge. The plane $\phi = 0$ is grounded and the plane $\phi = \alpha$ is at 100 V. Find V and E in the region between them, and evaluate E at r = 2 cm.

STEP-BY-STEP SOLUTION

1. The boundaries are surfaces of constant φ, so V depends only on φ: $\frac{1}{r^2}\frac{d^2V}{d\phi^2} = 0 \Rightarrow V = C_1\phi + C_2$.

2. $V(0) = 0 \Rightarrow C_2 = 0$. $V(\alpha) = V_0 \Rightarrow C_1 = V_0/\alpha$. So $V = \frac{V_0\phi}{\alpha}$.

3. $\mathbf{E} = -\mathbf{a}_\phi\frac{1}{r}\frac{\partial V}{\partial \phi} = -\mathbf{a}_\phi\frac{V_0}{\alpha r}$. **Don't forget the $1/r$.**

4. α must be in **radians**: $30^\circ = \pi/6 = 0.5236$. At r = 0.02 m: $|E| = \frac{100}{0.5236 \times 0.02} = 9549$ V/m.

Answer: $V = \frac{600}{\pi}\phi$ V, $\mathbf{E} = -\mathbf{a}_\phi\frac{100}{0.5236\,r}$, |E| ≈ 9.55 kV/m at 2 cm.

Q20. POSSIBLE

A conducting cone of half-angle α (apex at the origin, axis along z) is held at V_0 and insulated from a grounded conducting plane at $\theta = \pi/2$. Find V and E between them.

STEP-BY-STEP SOLUTION

1. Boundaries are θ = constant, so V = V(θ): $\frac{1}{R^2 \sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{dV}{d\theta}\right) = 0$.
2. $\sin\theta \frac{dV}{d\theta} = C_1 \Rightarrow \frac{dV}{d\theta} = \frac{C_1}{\sin\theta} \Rightarrow V = C_1 \ln\left(\tan\frac{\theta}{2}\right) + C_2$.
3. $V(\pi/2) = 0$: $\tan(\pi/4) = 1$ and $\ln 1 = 0$, so $C_2 = 0$.
4. $V(\alpha) = V_0 \Rightarrow C_1 = \frac{V_0}{\ln\tan(\alpha/2)}$. So $V = V_0 \frac{\ln\tan(\theta/2)}{\ln\tan(\alpha/2)}$.
5. $\mathbf{E} = -\mathbf{a}_\theta \frac{1}{R}\frac{dV}{d\theta} = -\mathbf{a}_\theta \frac{V_0}{R\sin\theta \ln\tan(\alpha/2)}$. Since $\alpha < \pi/2$, $\ln\tan(\alpha/2) < 0$, so E actually points in $+\mathbf{a}_\theta$: from the cone toward the plane, as expected.

Answer: $V = V_0 \frac{\ln\tan(\theta/2)}{\ln\tan(\alpha/2)}$, $\mathbf{E} = -\mathbf{a}_\theta \frac{V_0}{R\sin\theta \ln\tan(\alpha/2)}$.

Q21. ADVANCED, POSSIBLE

Two grounded, semi-infinite parallel conducting planes lie at $y = 0$ and $y = b$ for $x > 0$. The end $x = 0$ is closed by a conducting strip, insulated from the planes, held at V_0. Find V(x, y) inside the slot.

STEP-BY-STEP SOLUTION

1. **Boundary conditions** (no z dependence, so 2-D): (a) V(x, 0) = 0, (b) V(x, b) = 0, (c) V → 0 as x → ∞, (d) V(0, y) = V_0.
2. **Separate:** V = X(x)Y(y), so $X''/X = -Y''/Y = k^2$.
3. **Choose the forms.** In y, V must be zero at y = 0 and y = b, which needs a sine: $Y = \sin ky$. In x, V must decay, so $X = e^{-kx}$ (we discard e^{+kx} because it blows up).
4. **Apply (b):** $\sin kb = 0 \Rightarrow k = \frac{n\pi}{b}$, n = 1, 2, 3, …
5. **Superpose** (Laplace is linear): $V = \sum_n C_n e^{-n\pi x/b} \sin\frac{n\pi y}{b}$.
6. **Apply (d)** at x = 0: $V_0 = \sum C_n \sin\frac{n\pi y}{b}$ for 0 < y < b. This is a Fourier sine series. Multiply by $\sin\frac{m\pi y}{b}$ and integrate from 0 to b. Orthogonality leaves $C_m \frac{b}{2} = V_0 \int_0^b \sin\frac{m\pi y}{b} dy = \frac{V_0 b}{m\pi}(1 - \cos m\pi)$.
7. $C_n = \frac{2V_0}{n\pi}(1 - \cos n\pi)$. That gives $\frac{4V_0}{n\pi}$ for odd n and 0 for even n.

Answer: $V(x,y) = \dfrac{4V_0}{\pi} \displaystyle\sum_{n=1,3,5,\ldots} \frac{1}{n}\, e^{-n\pi x/b} \sin \frac{n\pi y}{b}.$

Interpretation: each Fourier mode decays exponentially into the slot, and higher modes decay faster. Far inside, only the n = 1 term matters: $V \approx \frac{4V_0}{\pi} e^{-\pi x/b} \sin \frac{\pi y}{b}$.

Q22. THEORY, VERY LIKELY

(a) State the uniqueness theorem and explain its importance. (b) Show that a function satisfying Laplace's equation cannot have a maximum inside a charge-free region. (c) Explain why Laplace's equation, not Poisson's, governs the region between the plates of a charged capacitor.

STEP-BY-STEP SOLUTION

(a) "A solution of Poisson's (or Laplace's) equation that satisfies the given boundary conditions is unique." Its importance: any solution, however it was found (guessing, images, separation of variables), is *the* solution if it satisfies the equation and the boundary conditions. Give the proof sketch from Section 6: $V_d = V_1 - V_2$ gives $\int |\nabla V_d|^2 dv = 0$, so $V_d = 0$.

(b) Suppose V had a maximum at a point P. At a maximum, every second derivative is ≤ 0, and for a strict maximum at least one is < 0. Then $\nabla^2 V = V_{xx} + V_{yy} + V_{zz} < 0$, which contradicts $\nabla^2 V = 0$. Equivalently, by the mean-value property V(P) equals the average over a small sphere around P, so it can't exceed every surrounding value. Therefore the extremes lie on the boundary.

(c) The capacitor's charges are *surface* charges on the conductors, which are the boundaries. The volume between the plates has $\rho = 0$, so $\nabla^2 V = 0$ there. The surface charges enter only through the boundary conditions (V given on each plate) and are recovered afterward via $\rho_s = \epsilon E_n$.

11 Rapid-fire concept checks

Cover the right-hand column and answer aloud.

#	Question	Answer
1	Units of energy density?	J/m³ (same as N/m², pressure)
2	Can $W_e = \frac{1}{2}\sum Q_k V_k$ be negative?	Yes, when there are opposite charges. It is interaction energy only.
3	Does $\frac{1}{2}\int \rho V \, dv$ include self-energy?	Yes
4	Self-energy of a point charge?	Infinite
5	E doubled, effect on w_e?	×4
6	Force formula with Q constant? With V constant?	$-\nabla W_e$; $+\nabla W_e$
7	Direction of electrostatic pressure on a conductor?	Outward, normal to the surface, whatever the charge sign
8	Poisson's equation?	$\nabla^2 V = -\rho/\epsilon$
9	Condition for Poisson to reduce to Laplace?	$\rho = 0$ (charge-free, simple medium)
10	∇^2 means?	Divergence of the gradient, $\nabla \cdot \nabla$
11	Why must ε be constant?	To take it out of $\nabla \cdot (\epsilon \nabla V)$
12	General V between coaxial cylinders?	$C_1 \ln r + C_2$
13	General V between concentric spheres?	$C_1/R + C_2$
14	V between two planes meeting at an angle?	$C_1\phi + C_2$
15	How many boundary conditions does a 1-D Laplace problem need?	Two (second-order ODE, two constants)
16	Can V have a minimum in a charge-free region?	No: extremes lie only on the boundaries
17	Theorem that justifies the method of images?	Uniqueness theorem
18	Dirichlet vs Neumann?	V specified vs ∂V/∂n specified
19	Child–Langmuir law?	$J \propto V_0^{3/2}/d^2$; $V \propto y^{4/3}$
20	Energy in a capacitor, three forms?	$\frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2/2C$

21	Where is E largest in a coax?	At the inner conductor ($E \propto 1/r$)
22	A dielectric slab is partly inserted into a charged, isolated capacitor. Is it pulled in or pushed out?	Pulled in: $W = Q^2/2C$ decreases as C increases
23	Separated solutions in x for V = 0 on two parallel walls?	sin / cos. In the open direction, exponentials or sinh/cosh.
24	1 eV in joules?	1.6×10^{-19} J

FINAL EXAM-HALL CHECKLIST

- Convert cm^2 to m^2 ($\times 10^{-4}$), mm to m, and degrees to radians.
- Write W in terms of whatever is held constant (Q or V).
- Include the scale factors in the gradient: $\frac{1}{r}\partial_\phi$, $\frac{1}{R}\partial_\theta$.
- Check that E points from high V to low V.
- Check that the boundary conditions are satisfied by your final V. That is all uniqueness requires.

Good luck. You've got this.